Sidewalk number stories

How a Sidewalk Is Made and How It Works

Name _____ Date _____

Use numbers or a drawing to share your thinking.

1. Little counts 36 tiny stones on one chip of concrete and 29 on another chip. How many stones is that in all?

36 + 29 = _____

Show your thinking.

2. A long sidewalk has 82 squares, and 47 of them have been swept. How many squares are left to sweep?

82 − 47 = _____

Show your thinking.

3. Big puts 45 pebbles in one jar and Little puts 38 pebbles in another jar. How many pebbles are in both jars?

45 + 38 = _____

Show your thinking.

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4. A sample tray holds 71 pebbles. Little presses 28 of them into wet paste. How many pebbles are left in the tray?

71 − 28 = _____

Show your thinking.

5. A new slab is kept damp for 7 days. It then has 21 more days to keep getting stronger. How many days is that in all?

7 + 21 = _____

Show your thinking.

6. A new path needs 72 grooves. Workers have pressed 48 so far. How many grooves are left to press?

72 − 48 = _____

Show your thinking.

Four ingredients

A sidewalk is concrete: gravel, sand, gray cement powder and water mixed together. Cement is only one ingredient, not the whole thing.

Paste that glues

Cement and water make a gray paste. It coats every stone and grain of sand and, once hard, holds them all tightly together.

Pour, spread, wait

Wet concrete is poured into a wooden form and smoothed. Cement and water then react and link together, so it hardens, even under water.

Curing takes days

A new slab is kept damp for about a week while the reaction goes on. This is curing. Most strength builds in the first month.

A line for cracks

Concrete shrinks as it hardens and can crack. Grooves make thin lines in the slab, so cracks tend to follow them in neat, straight lines.

Strong when squeezed

Concrete is strong when squeezed but weak when bent. Packed gravel under the slab keeps it level so the weight spreads out.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

How a Sidewalk Is Made and How It Works

Explain a sidewalk in your words

How a Sidewalk Is Made and How It Works · Curiosity day

Name _____ Date _____

Choose one clue. Say your idea first, then draw or write it.



Paste that glues



Poured into a frame



A line for cracks

Draw one square of sidewalk and add a small arrow to the groove.

concrete · cement · form · chemical reaction · curing · groove · because · so

Cement and water make a paste that … what does the paste hold?

After it is poured, the concrete … how does it turn hard?

A groove helps because … where does a crack go?

Does your sentence tell what happens and use because or so to explain why?